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EXPERIMENT NO:03

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Program:

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
outlook = ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain',  
           'Overcast', 'Sunny', 'Sunny', 'Rain', 'Sunny',  
           'Overcast', 'Overcast', 'Rain']
```

```
temperature = ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool',  
              'Cool', 'Mild', 'Cool', 'Mild', 'Mild',  
              'Mild', 'Hot', 'Mild']
```

```
play = ['No', 'No', 'Yes', 'Yes', 'Yes', 'No',  
       'Yes', 'No', 'Yes', 'Yes', 'Yes',  
       'Yes', 'Yes', 'No']
```

```
le1 = LabelEncoder()
```

```
le2 = LabelEncoder()
```

```
le3 = LabelEncoder()
```

```
outlook_encoded = le1.fit_transform(outlook)
```

```
temperature_encoded = le2.fit_transform(temperature)
```

```
play_encoded = le3.fit_transform(play)
```

```
X = list(zip(outlook_encoded, temperature_encoded))
```

```
y = play_encoded
```

```
model = DecisionTreeClassifier(criterion='entropy')
```

```
model.fit(X, y)
```

```
new_sample = [[le1.transform(['Sunny'])[0], le2.transform(['Cool'])[0]]]
```

```
prediction = model.predict(new_sample)
```

```
print("New Sample: Outlook = Sunny, Temperature = Cool")
```

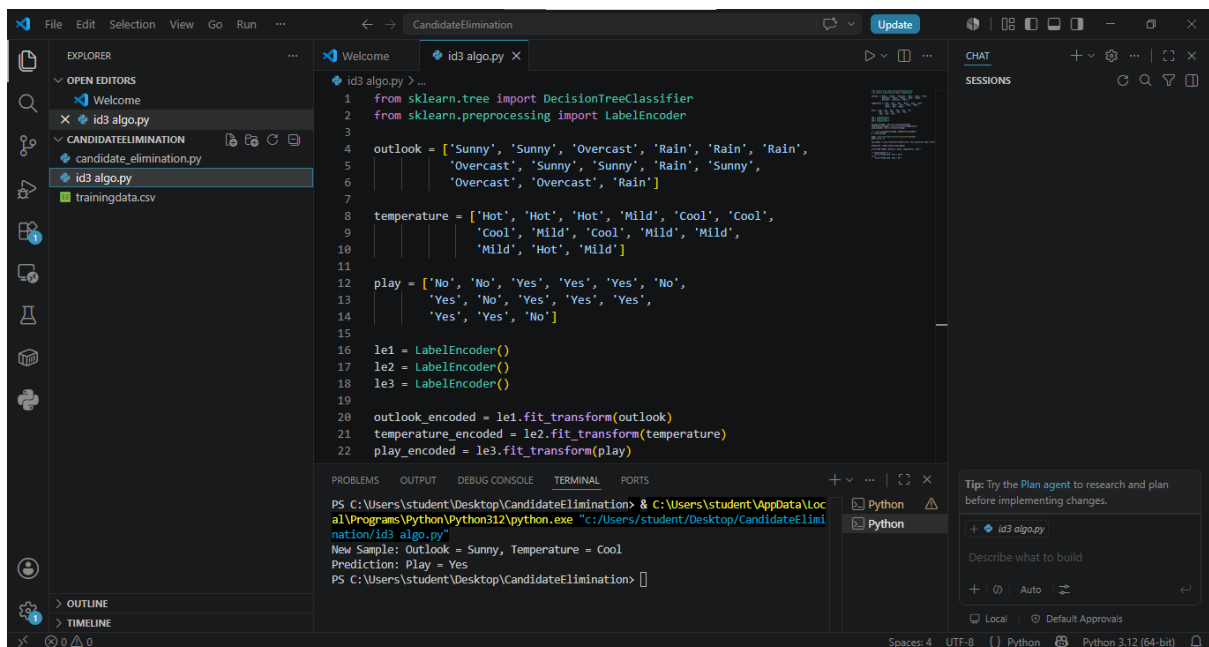
```
if prediction[0] == 1:
```

```
    print("Prediction: Play = Yes")
```

```
else:
```

```
    print("Prediction: Play = No")
```

output:



```
File Edit Selection View Go Run ...
CandidateElimination
id3 algo.py X
1 from sklearn.tree import DecisionTreeClassifier
2 from sklearn.preprocessing import LabelEncoder
3
4 outlook = ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain',
5           'Overcast', 'Sunny', 'Sunny', 'Rain', 'Sunny',
6           'Overcast', 'Overcast', 'Rain']
7
8 temperature = ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool',
9              'Cool', 'Mild', 'Cool', 'Mild', 'Mild',
10             'Mild', 'Hot', 'Mild']
11
12 play = ['No', 'No', 'Yes', 'Yes', 'Yes', 'No',
13        'Yes', 'No', 'Yes', 'Yes', 'Yes',
14        'Yes', 'Yes', 'No']
15
16 le1 = LabelEncoder()
17 le2 = LabelEncoder()
18 le3 = LabelEncoder()
19
20 outlook_encoded = le1.fit_transform(outlook)
21 temperature_encoded = le2.fit_transform(temperature)
22 play_encoded = le3.fit_transform(play)
23
24 new_sample = [[le1.transform(['Sunny'])[0], le2.transform(['Cool'])[0]]]
25
26 prediction = model.predict(new_sample)
27
28 print("New Sample: Outlook = Sunny, Temperature = Cool")
29
30 if prediction[0] == 1:
31     print("Prediction: Play = Yes")
32 else:
33     print("Prediction: Play = No")
34
35 # Output:
36 # New Sample: Outlook = Sunny, Temperature = Cool
37 # Prediction: Play = Yes
```

PS C:\Users\student\Desktop\CandidateElimination> & C:\Users\student\AppData\Local\Programs\Python\Python312\python.exe "c:/Users/student/Desktop/CandidateElimination/id3 algo.py"

New Sample: Outlook = Sunny, Temperature = Cool

Prediction: Play = Yes

PS C:\Users\student\Desktop\CandidateElimination>